

HW1 SOLUTION

1)

a) We will use induction . Let $c_1=1$ and $c_2 = 5$ and $n_0 = 10$. Our inequality becomes

$$n^3 \leq 4n^3 - 5n^2 + 7n \leq 5n^2$$

Lets start with the left side :

$$0 \leq 3n^3 - 5n^2 + 7n$$

that is true for $n=10$. So the base case for induction satisfied. Now assume the above inequality is true for k where $k \geq 10$. Now we have to show the inequality is also true for $k+1$. So we assume

$$0 \leq 3k^3 - 5k^2 + 7k$$

is true. And want to show

$$0 \leq 3(k+1)^3 - 5(k+1)^2 + 7(k+1)$$

is also true. Which means we need to show

$$0 \leq (3k^3 - 5k^2 + 7k) + (9k^2 - k + 5)$$

is true. Which means just showing

$$0 \leq (9k^2 - k + 5)$$

will be enough. Derivative of it is $(18k - 1)$ which means it is increasing after $\frac{1}{18}$. We assumed that $k > 10$ at the beginning that means the inequality is true for $k+1$ also

Now let's continue with the right side.

$$0 \leq k^3 + 5k^2 - 7k$$

is true for $k=10$. Assume it is true for some $k \geq 10$. Now we need to show it is also true for $(k+1)$. Which means

$$0 \leq (k^3 + 5k^2 - 7k) + (3k^2 + 13k - 1)$$

should be true. Proving

$$0 \leq (3k^2 + 13k - 1)$$

will be enough. Derivative of the left side is $6k + 13$ which is true for all $k > 0$. And we assumed k is greater than 10. This completes the proof.

b) Let $c=4$ and $n=17$. We need to show

$$3n^3 + 4n - 5 \leq 4n^2 - 12n + 4$$

is true. Which means we need to show $0 \leq n^2 - 16n + 9$. Which is $0 \leq (n-8)^2 - 55$ which is true for all $n > 16$. This completes the proof.

c) $\log(n)^3 < n$ for all $n > 10$. (Actually we also need to show that.) That means $5n \log(n)^3 < 5n^2$. So $3n^2 + 5n \log(n)^3 + 9n < 8n^2 + 9n < 17n^2$. Which means choosing $c=17$ and $n=1$ satisfies the condition

2)

a) $-1 \leq \sin \pi y \leq 1$ and $0 \leq (\sin \pi y)^2 \leq 1$. So $y^2 + 1$ grows faster.

b) We will use L'Hopital. $\lim_{n \rightarrow \infty} \frac{x}{\ln x}$. Differentiating both we get $\lim_{n \rightarrow \infty} \frac{x}{2 \ln x}$ and next $\lim_{n \rightarrow \infty} \frac{x}{2}$. Which means x grows faster.

c) $\lim_{n \rightarrow \infty} \frac{\arctan(x)}{\frac{1}{\arctan \frac{1}{x}}} = \lim_{n \rightarrow \infty} \arctan(x) \dots \lim_{n \rightarrow \infty} \frac{1}{\arctan \frac{1}{x}} = \frac{\pi}{2} \dots 0$. Which means $\frac{1}{\arctan \frac{1}{x}}$ grows faster.

3) ($k=1; k \leq j; k=k*3$) runs $\log_3 j$ times. So the code snippet loops

$$\sum_{i=1}^{2n} \sum_{j=1}^i \log_3 j$$

times in total. Which is equal to

$$\sum_{i=1}^{2n} \log_3 1 + \log_3 2 + \log_3 3 + \dots + \log_3 i - 1 + \log_3 i$$

. Which is equal to

$$\sum_{i=1}^{2n} \log_3 i!$$

. Now we know that

$$\log(n!) = \log(1) + \log(2) + \dots + \log(n) \leq \log(n) + \log(n) + \dots + \log(n) = n * \log(n)$$

. That is

$$\log(n!) \leq n * \log(n)$$

. So

$$\sum_{i=1}^{2n} \log_3 i! \leq \sum_{i=1}^{2n} i \log_3 i = 1 \cdot \log_3 1 + 2 \cdot \log_3 2 + \dots + (2n-1) \cdot \log_3 (2n-1) + 2n \cdot \log_3 2n \leq 4n^2 \log_3 2n$$

. Which is $O(n^2 \log n)$.

4) We need to use integral here. We will make an approximation

5)

a) Find the smallest element at each iteration.

b) Below is the python code of the algorithm.

```
def partition(list, start, end):
    pivot = list[end]                # Partition around the last value
    bottom = start-1                 # Start outside the area to be partitioned
    top = end                         # Ditto

    done = 0
    while not done:                  # Until all elements are partitioned...

        while not done:              # Until we find an out of place element...
            bottom = bottom+1         # ... move the bottom up.

            if bottom == top:         # If we hit the top...
                done = 1              # ... we are done.
                break

            if list[bottom] > pivot:   # Is the bottom out of place?
                list[top] = list[bottom] # Then put it at the top...
                break                 # ... and start searching from the top.

        while not done:              # Until we find an out of place element...
            top = top-1               # ... move the top down.

            if top == bottom:         # If we hit the bottom...
                done = 1              # ... we are done.
                break

            if list[top] < pivot:      # Is the top out of place?
                list[bottom] = list[top] # Then put it at the bottom...
                break                 # ...and start searching from the bottom.

    list[top] = pivot                # Put the pivot in its place.
    return top                       # Return the split point
```

```
def FindKSmallest(A,k):
    start=0
    end=len(A)-1
    med=partition(A,start,end)

    while(med!=k):
        if med>k:
            end=med-1
            med=partition(A,start,end)
        else:
            start=med+1
            med=partition(A,start,end)

    return A[0:k]
```